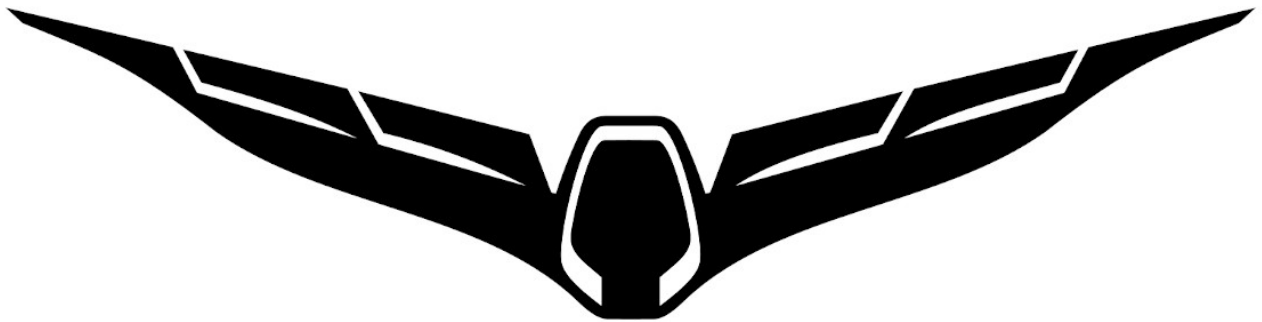


Thunderbird



**A customized firmware for Yuneec Typhoon H
based on PX4 Autopilot
by Toni Rosendahl**

User manual



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Brief description



"**Thunderbird**" is a customized firmware for Yuneec Typhoon H (aka H480) based on **PX4 Autopilot**. It's also a synonym for a Typhoon H with this PX4 based firmware.

With Thunderbird firmware you are able to use a lot of advanced features of PX4 Autopilot with stock hardware of Yuneec Typhoon H (H480) in combination with RC controller ST16. Controller ST16 will be used unchanged. That's why it is not part of this manual.

With the advanced software tool "[QGroundControl](#)" (QGC) you can configure, calibrate and control the Thunderbird.

QGroundControl allows you to setup and tune all parameters of the drone, but allows you also to do the wrong things. You should know what you do if you touch the parameters or settings.

The firmware was invented and developed by **Toni Rosendahl**.

First appearance (with a funny headline) was here:

<https://yuneecpilots.com/threads/typhoon-h-480-px4-v1-10-stability-issues.18205>

The whole project can be found at github: <https://github.com/tonirosendahl/Thunderbird>

Discover more about PX4 Autopilot: <https://docs.px4.io/>

The firmware is Open Source. **Please join the community and contribute!**

Important note:

You are responsible for all what you do. So, do only something if you know what you do.

Read the manuals for:

- PX4 Autopilot: <https://docs.px4.io/master/en/index.html>
- QGroundControl: <https://docs.qgroundcontrol.com/en/>

Operate Thunderbird in open area, away from people, vehicles and property.
Always be careful in what you do.

§ Follow local rules and regulations.

First steps

You have a complete new drone with different behavior compared to stock Typhoon H.

Get familiar with it and learn to fly the new drone. Be careful at the beginning, start in open and free area. Keep it away from trees.

You do not have to do anything on ST16 side. Stock settings there are OK.

It is highly recommended to read how to operate the PX4.

<https://docs.px4.io/master/en/flying/>

Power on/off

To power-on the drone hold the power button down for more than 8sec until the status-LED and LEDs on motor arms are on.

The drone keeps silence, it plays no sound.

To power-off the drone remove the battery. Powering off is currently not possible by power button.

Arming/disarming

Arming and disarming requires the **speed selector to be in FAST (rabbit) mode.**

The well known red button on the ST16 does NOT arm or disarm the drone.

We need a CSC command: Hold the throttle stick in the bottom right corner until motors starting.

Important note:

Tough it is not very likely the above described CSC command may disarm the drone when hold the bottom right position of the throttle stick too long. To spiral down right is not a good idea to bring the drone down.

Flying

In default settings, the upper position of the flight mode switch is **Stabilized** (see hints below), middle position is **Position Mode** (Typhoon with the GPS aka Angle Mode) and lower one is **Mission Mode**. RTH, Acro or Rattitude modes are not there by default. You can alter the switch configuration as you wish. There are two basic possibilities in QGroundControl that also could be combined:

- Switch Settings: Assign channels to flight modes, i.e. RTH to channel 10 (S2: Pan Mode),
- Flight Mode Settings: Assign flight modes to different values in channel 5 (A01) which can be set in ST16 Channel Settings.

Compared to stock Typhoon H you have to pay attention to following:

- In Position mode, the Thunderbird flies smoother. This is fine but you need more room because the "break path" is a bit longer than expected.
- In Stabilized mode, you have to maintain the altitude by yourself. Throttle stick means thrust in this case. If you pull the stick fully down, the Thunderbird comes down and I mean really down like a free falling stone.
- Before switch to Stabilized Mode or start in Stabilized Mode hold Throttle stick down. Middle Position means already middle thrust and the drone will climb immediately which may unexpected in this case.
- In all non-GNSS-assisted flight modes forget the ST16 screen. Keep both eyes on the drone.
- The Thunderbird is not a race drone! Keep in mind that its weight is ~2kg and it is fast. There is no such thing like FPV available due to latency of the video down-link.
- Do not switch from non-GNSS-assisted to GNSS-assisted flight modes (Position Mode or RTH) during high speed. It will report GPD error and will remain in non-GNSS-assisted mode. RTH is not possible in this case.
GPS module needs some time to get proper positioning. Let it hover in one position to switch the flight mode. Then wait for a couple of seconds before you proceed to fly.
- The green arrow on ST16's screen is **unreliable**. The arrow **is** there and visible, but please disregard it.

Important notes:

Sonar (OBS) is not supported and is always off! Skip warning messages regarding OBS on ST16.

ESC/Motor status is not monitored by the firmware. Eventually raised Motor error/Five-Rotor Mode messages are telemetry glitches and can be ignored.

The ST16 is not monitoring the voltage. It waits on the Voltage Warning Flags that never come from Thunderbird. **All pilots should be aware about that and monitor the voltage by themselves.** Don't drain the battery down to the minimum capacity. It is better to be conservative and bring it home with enough fuel in the tank.

At BAT_EMERGEN_THR (Default 7%) it will raise fail-save and land wherever it is without prior warning.

Sensor calibrations and settings

Sensor calibrations can only be done by QGroundControl. Connect the drone to QGC by long, flexible USB cable.

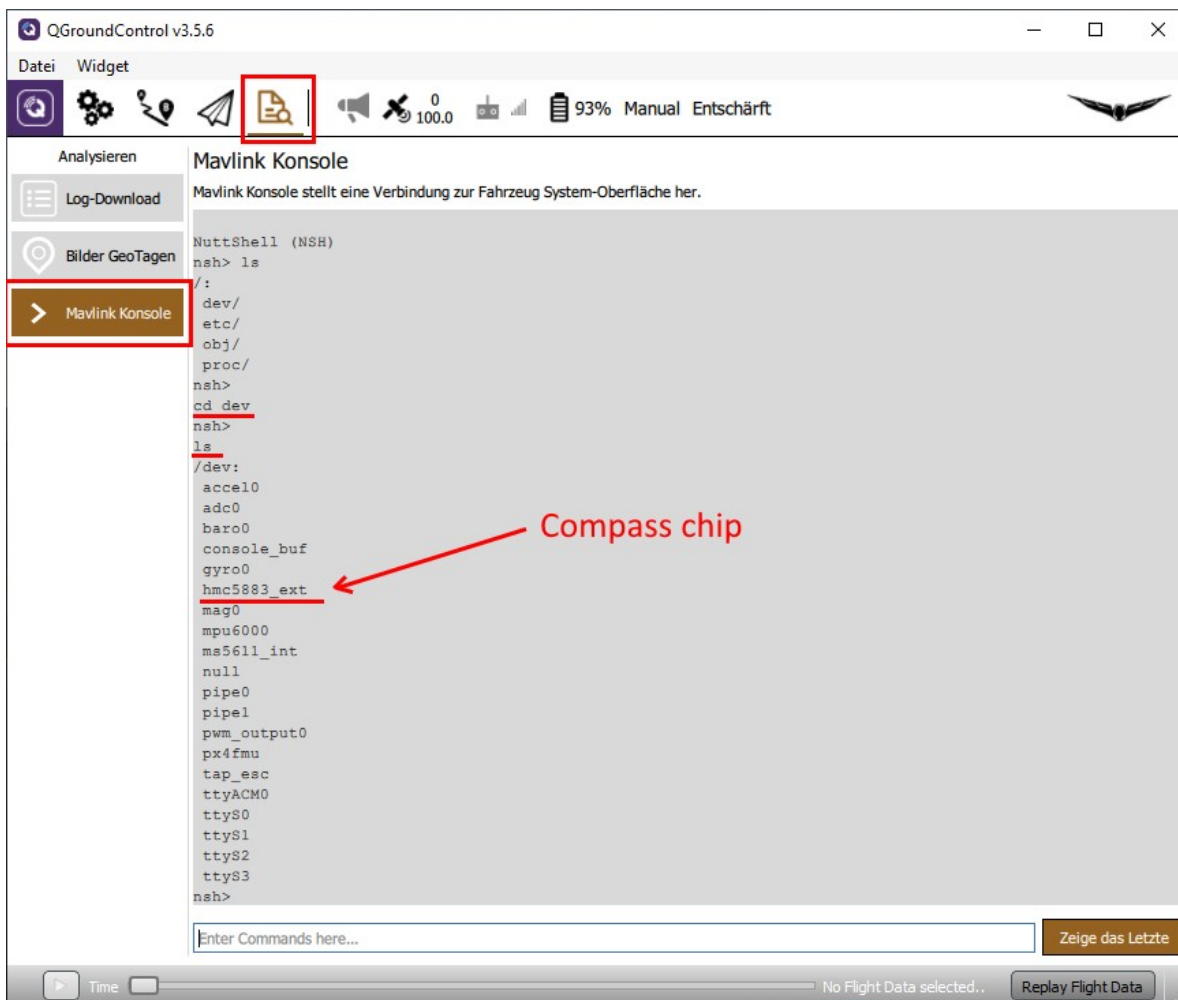
After changes changes, repairs and parameter file upload you need to check compass orientation and do a complete calibration.

Compass orientation settings

For Typhoon H exists two different compass chips. Older GPS boards have HMC5883, newer boards have IST8310 applied as compass chip (magnetometer). To find out which one you have, connect the drone to QGC and go to Mavlink console:

Type: `cd /dev`

then `ls` and you will get a list of drivers for the hardware.



There could be "hmc5883_ext" or "ist8310_ext" as compass chip.

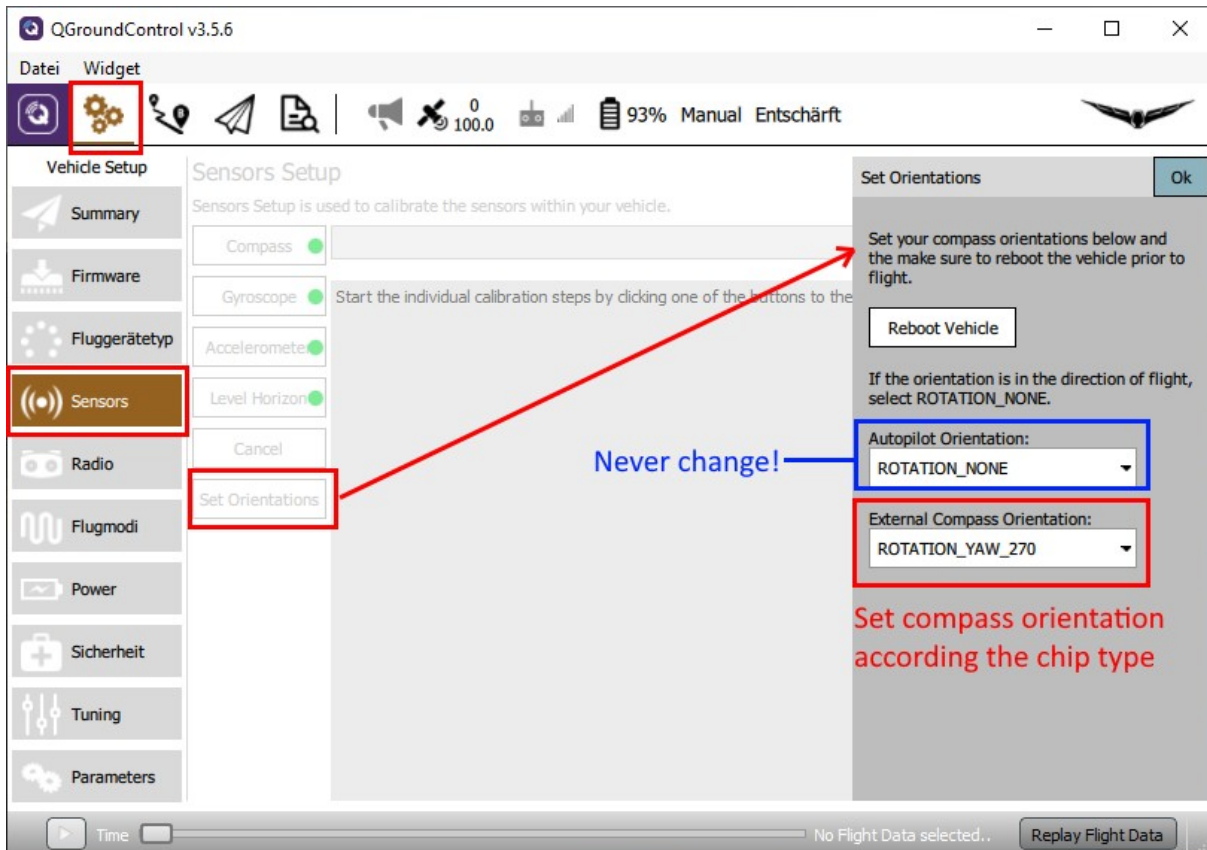
Note: If not all sensors are listed disconnect and reconnect to update the list.

Important: If you have the "hmc5883_ext", you have to change External Compass Orientation to "ROTATION_YAW_270".

For "ist8310_ext" it must be "ROTATION_YAW_180".

Never change Autopilot orientation. **It must be kept at "ROTATION_NONE"**.

Go to Settings > Sensors > Set Orientation



Then set External Compass Orientation to the correct value depending on you compass hardware. Save with "OK" and reboot the Thunderbird.

After changes do at least a compass calibration.

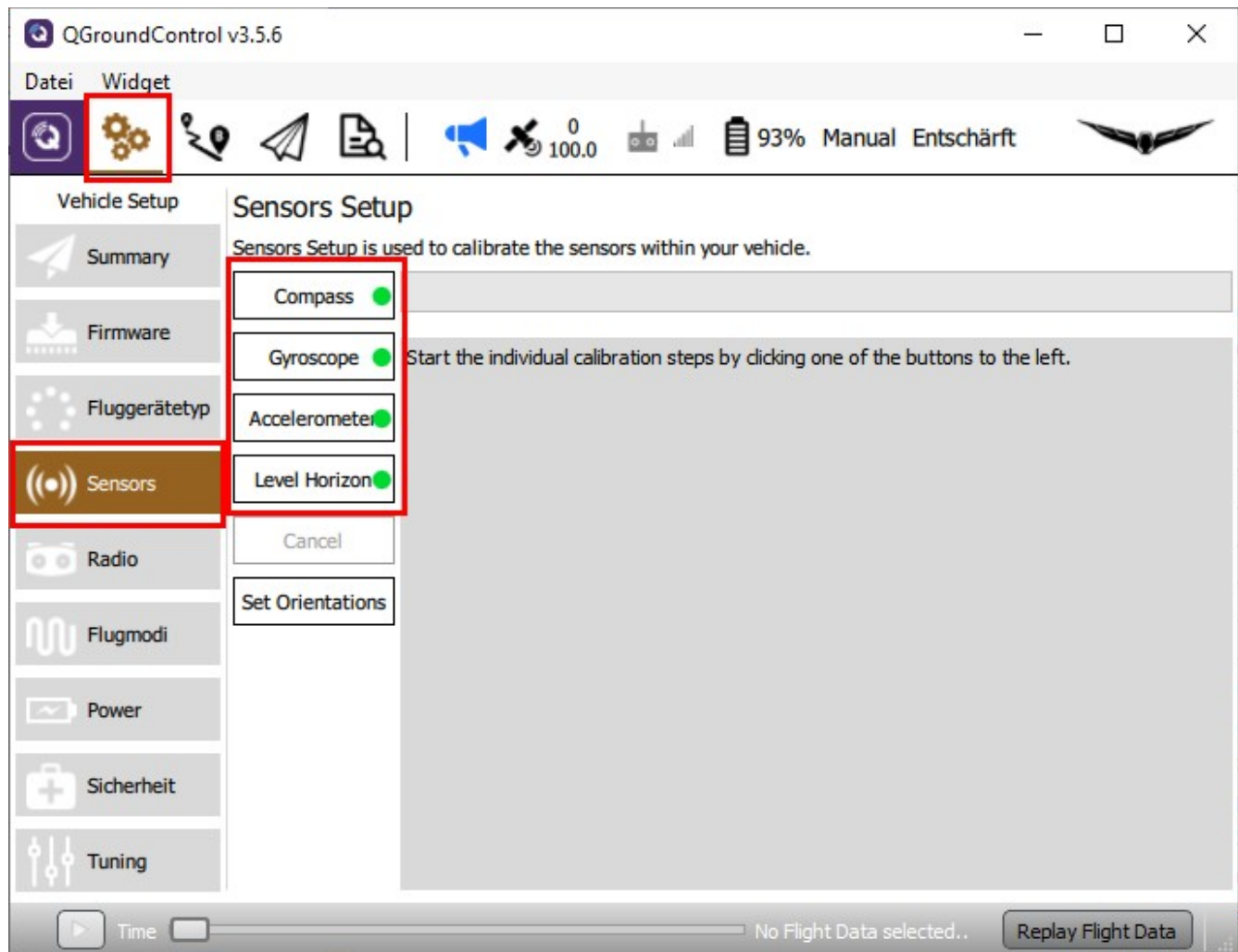
Read more: <https://docs.qgroundcontrol.com/en/SetupView/SetupView.html>

Calibration

Remove the camera.

Go to Settings > Sensors and follow the instructions on the screen for Compass, Gyroscope, Accelerometer and Level Horizon. **Do not try to calibrate ESC's.** It will not work and is not required. Flight battery must be well charged to allow proper calibration.

Prepare some fitting wooden blocks that keep the drone in backward, forward and lateral positions.



Learn more: https://docs.qgroundcontrol.com/en/SetupView/sensors_px4.html

Currently there is no "wireless" way to calibrate. **Be careful with the cable.** Best use a longer micro USB data cable.

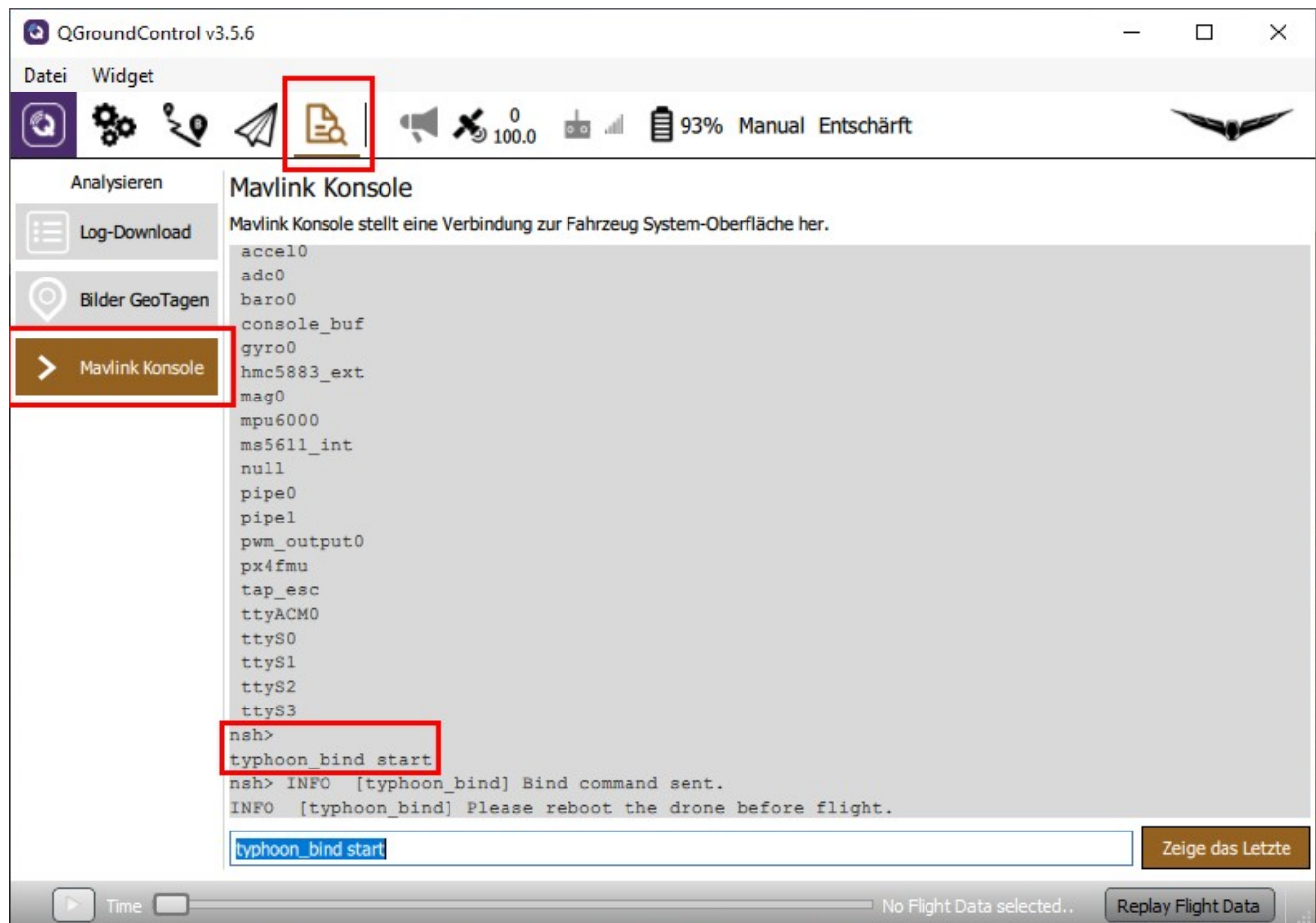
Note: If accelerometer BIAS is too high the accelerometer may need second calibration.

Bind procedure

Create a new model on ST16. Type is Typhoon H. Call it "My Thunderbird" or whatever you want. The bind process for camera CGO3+ is as usual.

To bind the 2.4GHz receiver the drone must be in bind mode. We have to use QGroundControl (QGC). Connect the drone to QGC by USB cable.

Go to Analyze Tools > Mavlink Console. Type: **typhoon_bind start** > Enter. The orange LED on SR24 receiver module starts blinking fast.



Then go to Bind menu of the ST16 as usual and tap on "Refresh". Select the receiver and tap on "Bind". Reboot the drone. Done!

Drone flight modes

Startup

Initialization

The GNSS and navigation systems are not ready. The home position is not set.

The drone can be forced to fly in non-GNSS modes but RTH will not be available during that flight.

GNSS-assisted flight modes are degraded to "Stabilize" with or without altitude hold. The GNSS-assisted modes will become available when the GNSS and estimator become ready. This will happen also during flight, but the RTH remains inoperable until you land and take-off again, as there is no home position set for this flight. The drone takes a minute or two to start up, to be fully mission capable. This never happens indoors, but it is perfectly flyable there.

Status LED: **GREEN blinking**. The ST16 shows "Acquiring" in GPS mode display and "Start" as a flight mode.

Ready

The drone is ready. It has full navigation capability and home position is set.

Status LED: **GREEN solid**. ST16 GPS status changes from "Acquiring" to "Ready" and flight mode display shows also "Ready". You are good to go, no need to wait any further.

GNSS assisted flight modes

Position mode

The Thunderbird flies like a stock Typhoon in Angle Mode.

Status LED: **PURPLE solid**, ST16 flight mode display: "Angle".

Note: All sensors and GNSS must be fully operational and perfectly calibrated for these modes.

Mission mode

The Thunderbird is fully automatic. This is the automated flight mode and easiest of them all since the drone flies itself from take-off to motor shutdown. If no mission is available it switches back to Position Mode.

Status LED: **PURPLE blinking**, ST16 flight mode display: "Waypoint".

Stabilized flight modes

Altitude mode

Stabilized mode with altitude hold and assisted throttle. Auto-throttle is active, throttle stick controls altitude.

The Thunderbird flies like a stock Typhoon in Angle Mode without GPS.

Status LED: **BLUE blinking**, ST16 flight mode display: "THR".

Stabilize mode

Throttle controls altitude. The Thunderbird controls thrust to maintain the altitude. If the Position mode was requested with no valid positioning solution, meaning you took off before the estimator and GNSS were ready, this is what you will get instead as a fallback.

This flight mode is not mapped to the mode selector switch by default at the standard parameter settings. But you can summon this by requesting a position mode from a drone that took off before it was ready. Same as "Stability" mode, known from Blade Chroma or Blade 350QX.

Status LED: **BLUE solid**, ST16 flight mode display: "THR".

Note: These modes do not require GPS magnetometer being operational. Indoor flight is possible and magnetic interference are no issue. These modes can be used to recover the drone from GPS/magnetometer issues, if you'll ever encounter any. These modes require active controlling and visual line of sight, but they are still rather easy to fly.

Do not switch back to GNSS-assisted modes or RTH if the drone is moving fast. Let it hover to do that.

Manual flight modes

Rattitude mode

"Assisted Acrobatics" enabled. Release the sticks and the drone *should* self-recover if there is enough altitude.

Status LED: ~~WHITE~~ **purple blinking**, ST16 flight mode display: "Rate".

Manual, Acro or Rate mode

Sticks control angular acceleration, no self-leveling, no braking, direct thrust control, absolutely zero assistance from the drone. The GPS, magnetometer, barometer and accelerometer are sleeping, only the gyro data is used. You are in control now.

Same as "Agility" mode, known from Blade Chroma or Blade 350QX.

Status LED: **WHITE solid**, ST16 flight mode display: "Rate".

Note: These modes are **not set** to the mode selector switch in my default parameter file. You have to enable these by yourself. The "Purple solid" is a **very difficult mode** to fly, but allows you do anything with the drone.

Be careful with Acro or Rattitude mode! First learn to fly Stabilized mode.

Fail-safe mode

Fail-safes are activated (RTH) or initiated by the Autopilot. In case of low battery or failed GPS the drone attempts auto-land.

Status LED: **RED blinking**.

Drone status LED codes

Drone status LED basic rules, regardless of color coding

1. When the LED is SOLID regardless of color, you have the full control.
2. When the LED is BLINKING, the Thunderbird is actively doing something (like assisting you somehow).
3. Than lighter the color, than more challenging and rewarding is it to fly.
4. Red light means malfunction, fail-safe mode or manual requested RTH.

RGB status LED details

Green: Startup

GREEN blinking: The GNSS and navigation systems are not ready and home position is not set. The ST16 shows "Acquiring" in GPS mode display and "Start" as a flight mode.

GREEN solid: The drone is **ready**. The green light will not appear when the drone is airborne. ST16 GPS status changes from "Acquiring" to "Ready" and flight mode display shows also "Ready". You are good to go, no need to wait any further.

Purple: GNSS-assisted flight

PURPLE blinking: Mission mode. ST16 flight mode display: "Waypoint".

PURPLE solid: Position mode. ST16 flight mode display: "Angle".

Blue: Stabilized flight

BLUE blinking: Altitude mode

BLUE solid: Stabilize mode. ST16 flight mode display: "THR".

White: Manual flight modes

~~WHITE~~ **purple blinking:** Rattitude mode or "Assisted Acrobatics".

ST16 flight mode display: "Rate".

~~WHITE~~ **solid:** Manual, Acro or Rate mode. ST16 flight mode display: "Rate" (not tested yet).

Red: Fail-safe, RTH, malfunction

RED blinking: The drone has degraded performance, but is airborne, still operational and attempts an automatic recovery. The LED blinks red when the fail-safes are activated (RTH) or the drone attempts auto-land with a failed GPS. It may change between red and previous color code for some time. However, it is recommended to bring it home at this point.

RED solid: Malfunction, arming is prohibited. ST16 flight mode display: "EMER".

Parameter settings

Parameter settings needs to be done with QGroundControl. Changed parameters are **not** stored in the model on ST16.

It is recommended to store last working parameter set into a file before you change something. Make a note what parameter you have changes and why (see Parameter change sheet at the end of this document). The stored parameter file is a backup and can be used to restore the whole parameter set in case of problems.

Change parameters

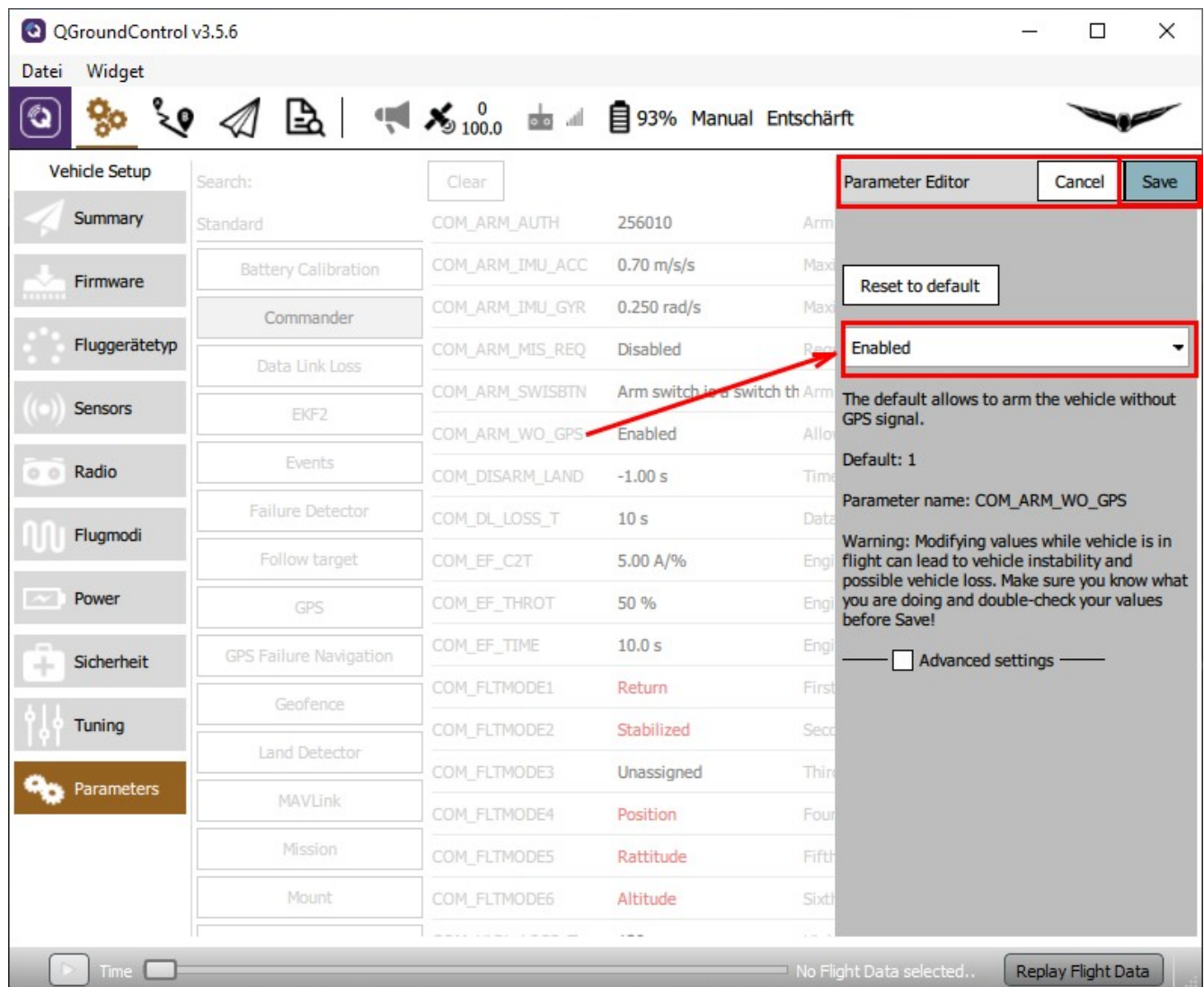
Power up the drone and connect it via USB to QGroundControl.

Go to Settings (icon with gearwheels) > Parameters. Parameter sets are grouped by its functionality. To find a parameter you can use the search function.

The screenshot shows the QGroundControl v3.5.6 interface. The top bar includes a search field, a 'Clear' button, and a 'Tools' button. The left sidebar contains various settings categories: Vehicle Setup, Summary, Firmware, Fluggerätetyp, Sensors, Radio, Flugmodi, Power, Sicherheit, Tuning, and Parameters (highlighted with a red box). The main area displays a list of parameters grouped by functionality. The 'Standard' group is selected, showing parameters like COM_ARM_AUTH, COM_ARM_IMU_ACC, COM_ARM_IMU_GYR, etc. The 'Parameters' category in the sidebar is highlighted with a red box.

Parameter	Value	Description
COM_ARM_AUTH	256010	Arm authorization parameters, this uint32_t will be set
COM_ARM_IMU_ACC	0.70 m/s/s	Maximum accelerometer inconsistency between IMU units
COM_ARM_IMU_GYR	0.250 rad/s	Maximum rate gyro inconsistency between IMU units
COM_ARM_MIS_REQ	Disabled	Require valid mission to arm
COM_ARM_SWISBTN	Arm switch is a switch th	Arm switch is only a button
COM_ARM_WO_GPS	Enabled	Allow arming without GPS
COM_DISARM_LAND	-1.00 s	Time-out for auto disarm after landing
COM_DL_LOSS_T	10 s	Datalink loss time threshold
COM_EF_C2T	5.00 A/%	Engine Failure Current/Throttle Threshold
COM_EF_THROT	50 %	Engine Failure Throttle Threshold
COM_EF_TIME	10.0 s	Engine Failure Time Threshold
COM_FLTMODE1	Return	First flightmode slot (1000-1160)
COM_FLTMODE2	Stabilized	Second flightmode slot (1160-1320)
COM_FLTMODE3	Unassigned	Third flightmode slot (1320-1480)
COM_FLTMODE4	Position	Fourth flightmode slot (1480-1640)
COM_FLTMODE5	Rattitude	Fifth flightmode slot (1640-1800)
COM_FLTMODE6	Altitude	Sixth flightmode slot (1800-2000)

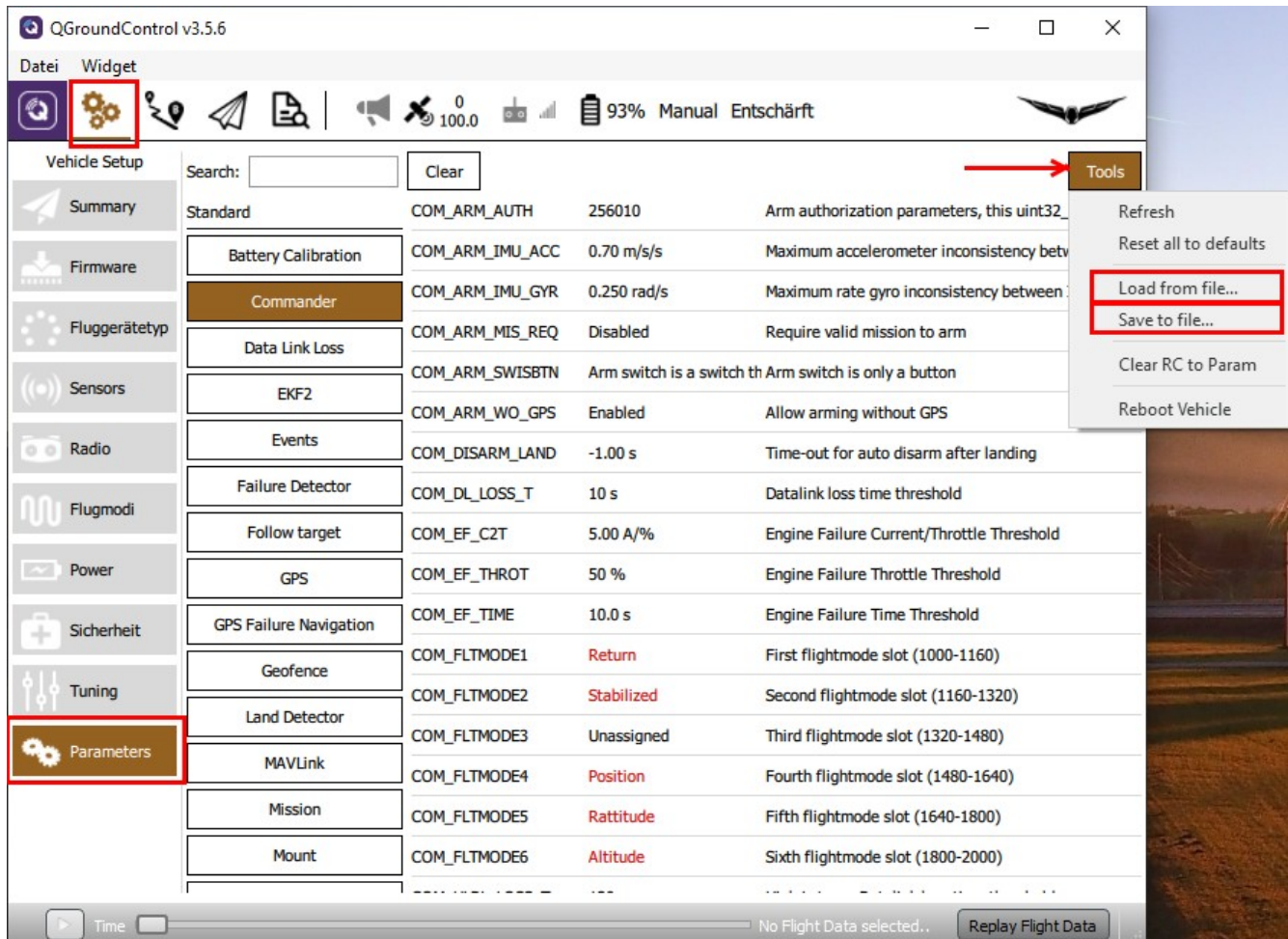
Click on the parameter you want to change. The Parameter editor appears on the right.



Change parameter (select from list or enter value) and Save. Some parameters require reboot of the drone.

Save parameter sets

Go to Settings (icon with gearwheels) > Parameters > Tools > Save to file... Enter useful path and file-name to remember what parameter set it was and save it.



Restore parameter from file

The parameter file from the last working configuration can be used to restore parameter settings in case of problems with the new one.

Go to Settings (icon with gearwheels) > Parameters > Tools > Load from file... Select parameter file with last working settings and upload it to the drone.

Reboot the drone to make the new parameter set happen. Calibration settings are part of the parameters and will be overwritten.

Check Compass orientation depending on compass chip and do proper calibration.

Firmware update procedure using Ubuntu LINUX

Copy following files in a separate directory:

flash_typhoon_bootloader
px_uploader.py
yuneec_typhoon_h.fw

Start script
Flash utility
The firmware itself

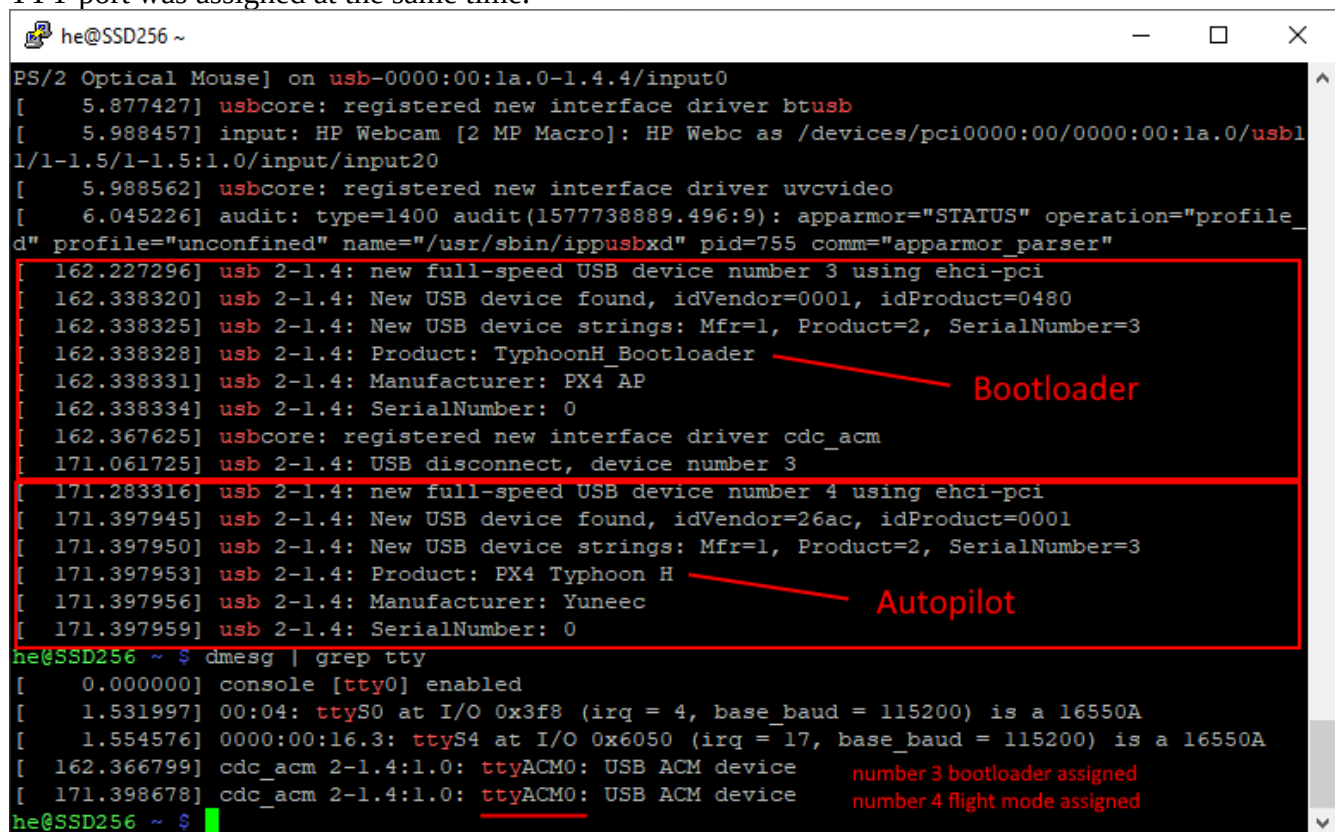
⚠ The file name for the firmware may change. "yuneec_typhoon_h.fw" is used here as example.

Find serial port

To find out what TTY port is used by the drone you have to disconnect and connect again the USB cable between drone and LINUX PC. Open a terminal and enter following commands:

```
dmesg | grep usb  
dmesg | grep tty
```

The first command shows which USB devices were detected. The second command shows which TTY port was assigned at the same time.



```
he@SSD256 ~  
PS/2 Optical Mouse] on usb-0000:00:1a.0-1.4.4/input0  
[ 5.877427] usbcore: registered new interface driver btusb  
[ 5.988457] input: HP Webcam [2 MP Macro]: HP Webc as /devices/pci0000:00/0000:00:1a.0/usb1  
1/1-1.5/1-1.5:1.0/input/input20  
[ 5.988562] usbcore: registered new interface driver uvcvideo  
[ 6.045226] audit: type=1400 audit(1577738889.496:9): apparmor="STATUS" operation="profile_  
d" profile="unconfined" name="/usr/sbin/ippusbxd" pid=755 comm="apparmor_parser"  
[ 162.227296] usb 2-1.4: new full-speed USB device number 3 using ehci-pci  
[ 162.338320] usb 2-1.4: New USB device found, idVendor=0001, idProduct=0480  
[ 162.338325] usb 2-1.4: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
[ 162.338328] usb 2-1.4: Product: TyphoonH_Bootloader  
[ 162.338331] usb 2-1.4: Manufacturer: PX4 AP  
[ 162.338334] usb 2-1.4: SerialNumber: 0  
[ 162.367625] usbcore: registered new interface driver cdc_acm  
[ 171.061725] usb 2-1.4: USB disconnect, device number 3  
[ 171.283316] usb 2-1.4: new full-speed USB device number 4 using ehci-pci  
[ 171.397945] usb 2-1.4: New USB device found, idVendor=26ac, idProduct=0001  
[ 171.397950] usb 2-1.4: New USB device strings: Mfr=1, Product=2, SerialNumber=3  
[ 171.397953] usb 2-1.4: Product: PX4 Typhoon H  
[ 171.397956] usb 2-1.4: Manufacturer: Yuneec  
[ 171.397959] usb 2-1.4: SerialNumber: 0  
he@SSD256 ~ $ dmesg | grep tty  
[ 0.000000] console [tty0] enabled  
[ 1.531997] 00:04: ttyS0 at I/O 0x3f8 (irq = 4, base_baud = 115200) is a 16550A  
[ 1.554576] 0000:00:16.3: ttyS4 at I/O 0x6050 (irq = 17, base_baud = 115200) is a 16550A  
[ 162.366799] cdc_acm 2-1.4:1.0: ttyACM0: USB ACM device  
[ 171.398678] cdc_acm 2-1.4:1.0: ttyACM0: USB ACM device  
he@SSD256 ~ $
```

Here we found, the port is "ttyACM0".

Also we can see that after power-on (USB connected again) the bootloader starts first and after a while the Autopilot starts. If Autopilot is running we cannot flash anymore. This is the reason why we have to

start the script for flashing prior to power-up the MCU board which must only be powered via USB. If we plug-in the USB cable when the script is already running we will be able to catch the bootloader.

Flashing

Now we edit the file "flash_typhoon_bootloader" to set the correct port. The file text should look like that:

```
python px_uploader.py --port /dev/tttyACM0 --force yuneec_typhoon_h.fw
```

In terminal, move to the directory where the three files are located and make the scripts executable:

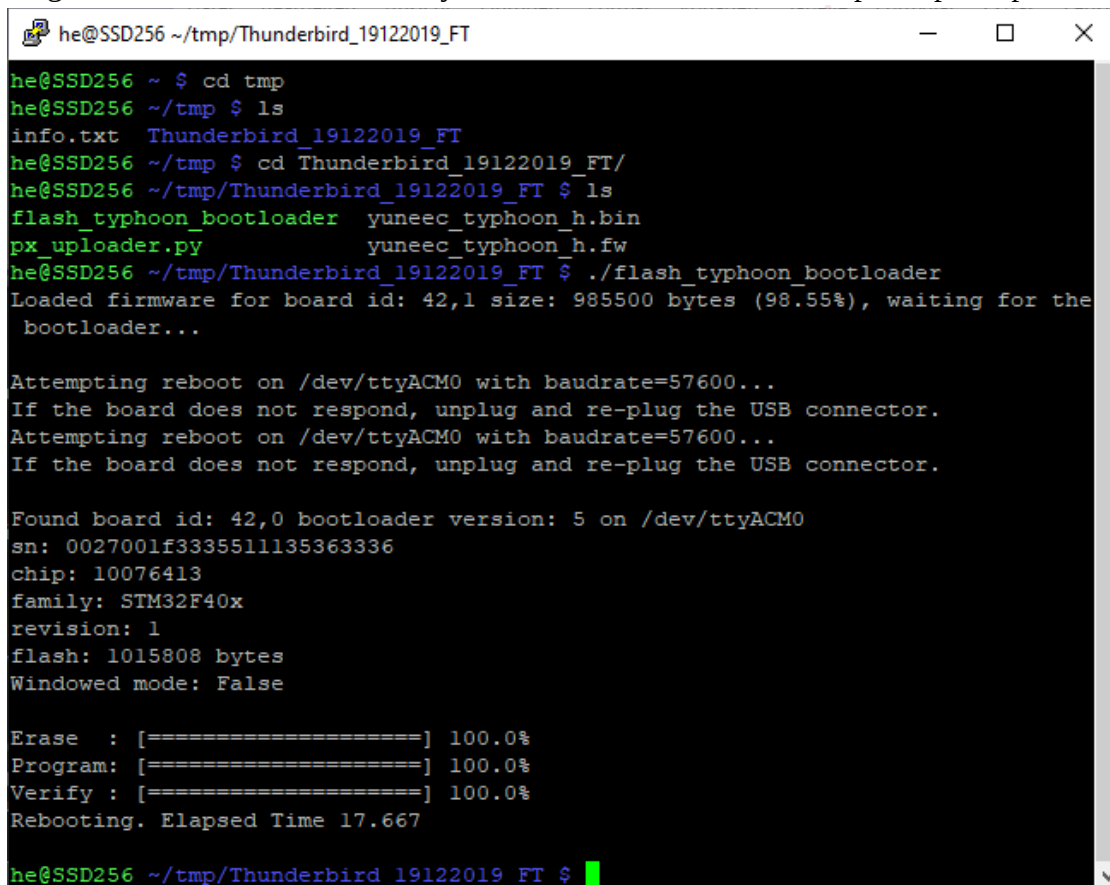
```
chmod +x ./flash_typhoon_bootloader  
chmod +x ./px_uploader.py
```

⚠️ **Make sure the USB connection is cut and no battery in the drone. (MCU-board powered off).**

Start firmware update script:

```
./flash_typhoon_bootloader
```

Plug in the USB cable immediately. Wait and observe terminal output. Update process will start soon...

A terminal window titled 'he@SSD256 ~/tmp/Thunderbird_19122019_FT' displays the execution of the 'flash_typhoon_bootloader' script. The script first checks for the presence of 'flash_typhoon_bootloader' and 'yuneec_typhoon_h.fw' files. It then attempts to reboot the board on '/dev/tttyACM0' with a baudrate of 57600. After two failed attempts, it successfully finds the board (id: 42,0) and displays its details: sn: 0027001f3335511135363336, chip: 10076413, family: STM32F40x, revision: 1, flash: 1015808 bytes, and Windowed mode: False. The script then proceeds to erase the flash (100.0%), program the new firmware (100.0%), and verify it (100.0%). The process concludes with a reboot, taking 17.667 seconds. The prompt returns to the user at the terminal path.

```
he@SSD256 ~ $ cd tmp  
he@SSD256 ~/tmp $ ls  
info.txt  Thunderbird_19122019_FT  
he@SSD256 ~/tmp $ cd Thunderbird_19122019_FT/  
he@SSD256 ~/tmp/Thunderbird_19122019_FT $ ls  
flash_typhoon_bootloader  yuneec_typhoon_h.bin  
px_uploader.py            yuneec_typhoon_h.fw  
he@SSD256 ~/tmp/Thunderbird_19122019_FT $ ./flash_typhoon_bootloader  
Loaded firmware for board id: 42,1 size: 985500 bytes (98.55%), waiting for the  
bootloader...  
  
Attempting reboot on /dev/tttyACM0 with baudrate=57600...  
If the board does not respond, unplug and re-plug the USB connector.  
Attempting reboot on /dev/tttyACM0 with baudrate=57600...  
If the board does not respond, unplug and re-plug the USB connector.  
  
Found board id: 42,0 bootloader version: 5 on /dev/tttyACM0  
sn: 0027001f3335511135363336  
chip: 10076413  
family: STM32F40x  
revision: 1  
flash: 1015808 bytes  
Windowed mode: False  
  
Erase : [=====] 100.0%  
Program: [=====] 100.0%  
Verify : [=====] 100.0%  
Rebooting. Elapsed Time 17.667  
he@SSD256 ~/tmp/Thunderbird_19122019_FT $
```

Reboot drone after firmware was flashed successfully. Done!
Do all checks and calibrations like for a new drone.

Firmware update procedure for Windows

Preparation

Copy following files in an own directory:

flash_typhoon_bootloader
px_uploader.py
yuneec_typhoon_h.fw

Start script
Flash utility
The firmware itself

⚠ The file name for the firmware may change. "yuneec_typhoon_h.fw" is used here as example.

Install Python for Windows if not yet done:

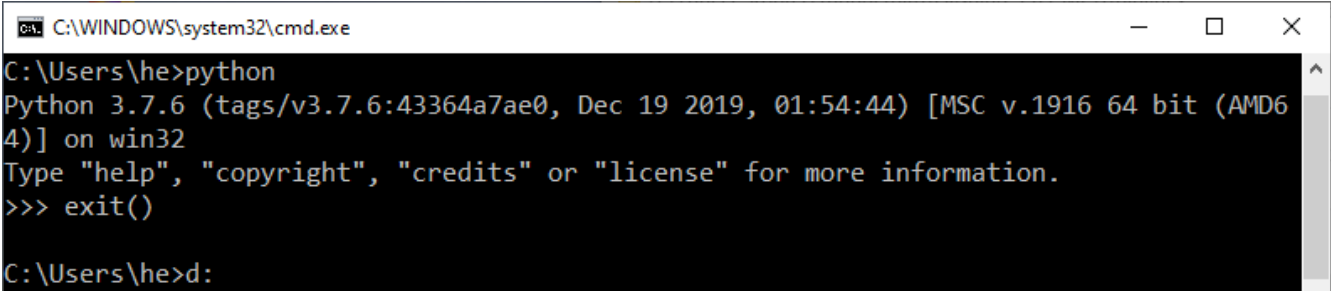
Windows key + R to open command field.
Type **cmd**

The Windows terminal will be opened. Keep it open, we need it all the time.

Type **python**

The Windows Store will be opened and offer Python installation. Follow the instructions to install Python.

If Python is correctly installed it should come up with its own command line >>>.



```
C:\WINDOWS\system32\cmd.exe
C:\Users\he>python
Python 3.7.6 (tags/v3.7.6:43364a7ae0, Dec 19 2019, 01:54:44) [MSC v.1916 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> exit()
C:\Users\he>d:
```

Leave it with **exit()**.

Then download module 'serial' for Python::

<https://pypi.org/project/pyserial/#files>

File name is "pyserial-3.5-py2.py3-none-any.whl", double click on it to install.

Python is installed now and we can run Python scripts like "px_uploader.py".

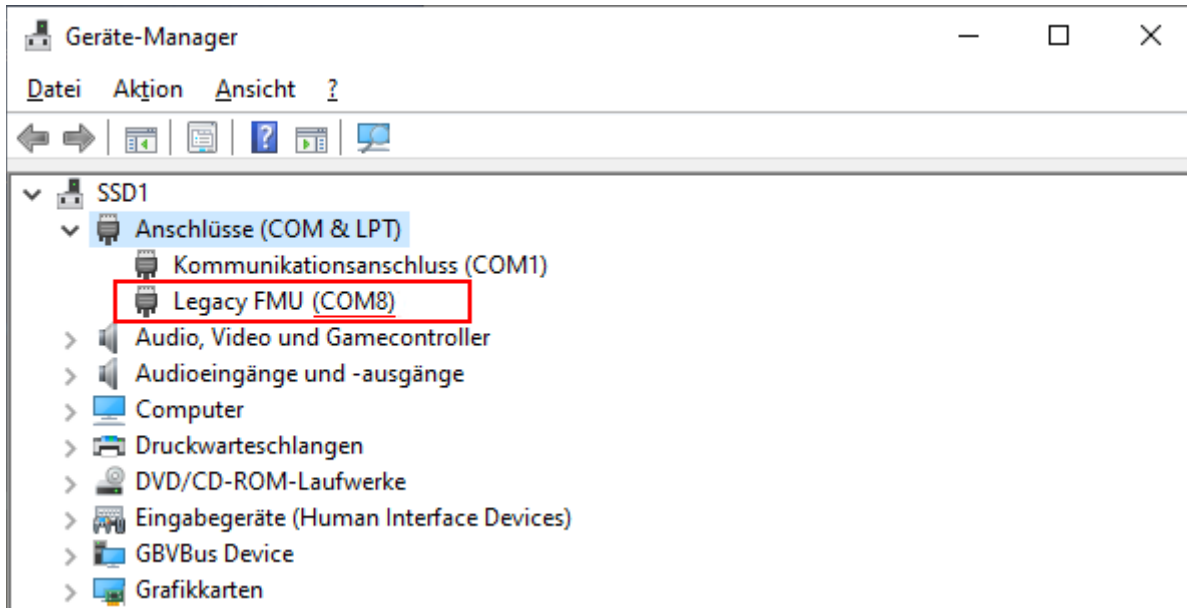
Find serial port

To find out what TTY port is used by the drone you have to check the Device Manager.

Windows key + R to open command field.

Type **devmgmt.msc**

The Device Manager will appear. Plug-in the USB cable to connect the drone with the PC. In node "Connections (COM & LPT)" an item "Legacy FMU" with a COM number appears.



Here we found, the port is "**COM8**".

Now we edit the file "flash_typhoon_bootloader" to set the correct port and save it as

"**flash_typhoon_bootloader.bat**" to make it executable. The file text should look like that:

```
python px_uploader.py --port COM8 --force "yuneec_typhoon_h.fw"
```

Flashing

Note: After power-on the bootloader starts first and after a while the flight controller starts. If flight controller is running we cannot flash anymore. This is the reason why we have to start the script for flashing prior to power-up the MCU board which is powered by flight battery. If we plug-in the USB cable when the script is already running we will be able to catch the bootloader.

⚠ **Make sure the USB connection is unplugged (MCU-board powered off).**

Start firmware update script:

flash_typhoon_bootloader.bat

Plug in the USB cable. Then hold the start button of the aircraft down until flashing is complete, disconnect and connect USB cable again. Wait and observe terminal output.

Update process will start soon ...

```
PS D:\temp\Thunderbird_170521_CD> .\flash_typhoon_bootloader.bat

D:\temp\Thunderbird_170521_CD>python px_uploader.py --port COM8 --force yuneec_typhoon_h.fw
Loaded firmware for board id: 42,1 size: 970768 bytes (97.08%), waiting for the bootloader...

Attempting reboot on COM8 with baudrate=57600...
If the board does not respond, unplug and re-plug the USB connector.

Found board id: 42,0 bootloader version: 5 on COM8
sn: 0038005b3335510537363336
chip: 10076413
family: b'STM32F40x'
revision: b'1'
flash: 1015808 bytes
Windowed mode: False

Erase   : [=====] 100.0%
Program: [=====] 100.0%
Verify  : [=====] 100.0%
Rebooting. Elapsed Time 16.328

PS D:\temp\Thunderbird_170521_CD> |
```

Done!

Reboot drone after firmware was flashed successfully.

Start QGroundControl and reload parameters (your parameter backup or the default parameter set that came with latest firmware).

Vehicle Setup > Parameters > Tools > Load from file... > Select the parameter file.

Reboot again.

Do all checks and calibrations like for a new drone.

Annex

Recommended settings

Parameter reference: https://docs.px4.io/v1.9.0/en/advanced_config/parameter_reference.html

Parameter	from	to	Description
MPC_Z_VEL_MAX_DN	1.000	3.000	Max. descent speed [m/s] as it was at H480
MPC_Z_VEL_MAX_UP	3.0	5.0	Max. ascent speed [m/s] as it was at H480
MC_RATT_TH	0.80	0.60	Threshold for Stabilized in Rattitude mode [%] for more Acro to get smoother rolls or loopings
COM_ARM_WO_GPS	Disabled (0)	Enabled (1)	For indoor flights. For GNSS-assisted flights wait for solid green LED status.
RTL_RETURN_ALT	30m	x	Keep default Return altitude or change it to a value x that fits better to your area.

Flight Modes:

QGroundControl > Flight Modes > Flight Mode Settings:

Mode Channel = Channel 5

Flight Mode 6: change from **Stabilized** to **Altitude**. Altitude Mode is the same as Typhoon H with GPS off.

QGroundControl > Flight Modes > Switch Settings:

Return Switch Channel: Change from Unassigned to **Channel 10**. In this case S2 Pan Mode lowest position is RTH. This switch overwrites all other flight modes.

Notes:

- Test changed flight modes without propellers before you use this settings in real flight sessions.
- All tuning measurements and parameter setting may influence flight behavior and security. Do it on your own risk.

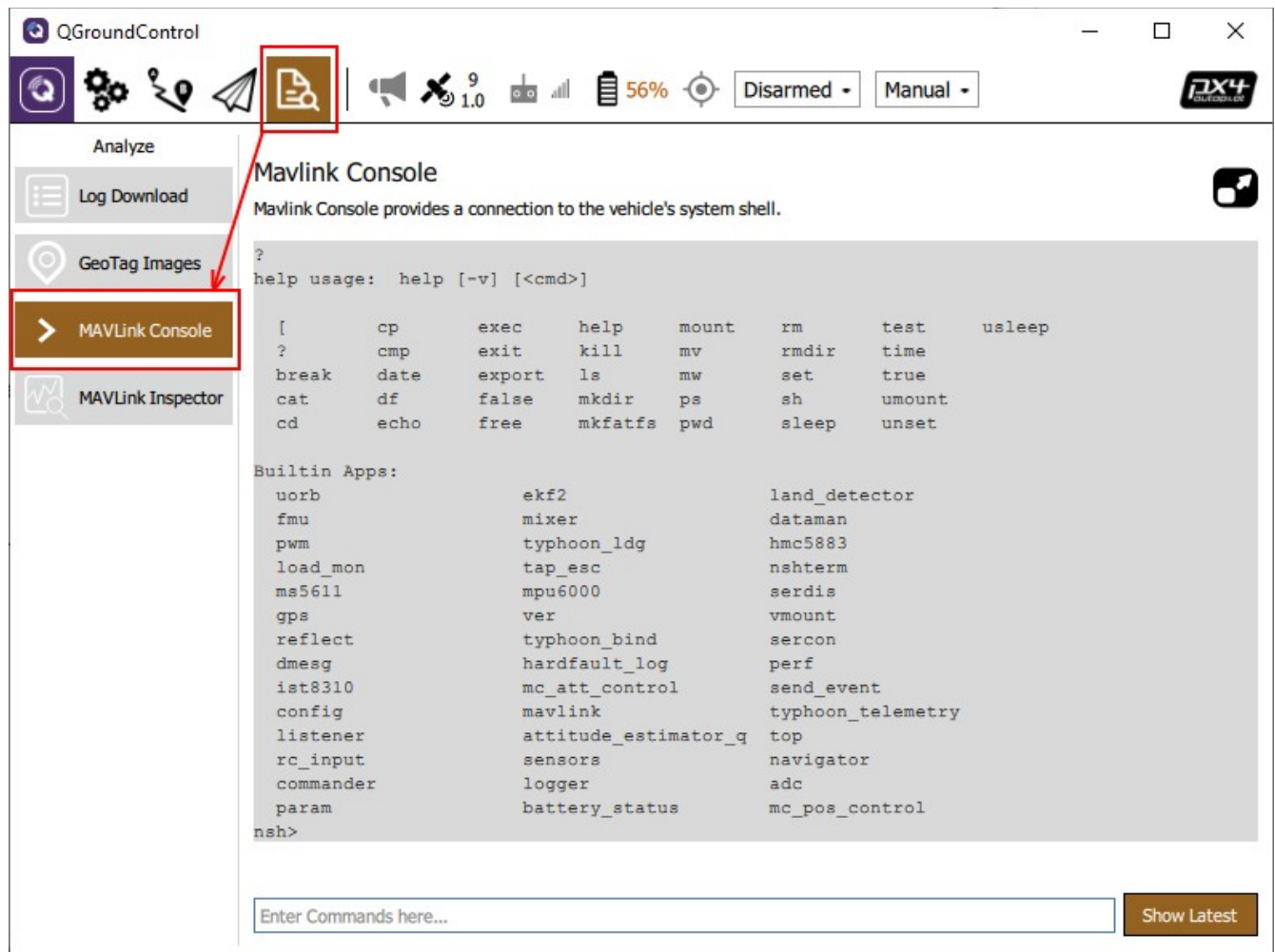
Some helpful MAVLink Console commands

The MAVLink Console opens a PX4 NSH shell to send commands and receive results when the aircraft is powered on and connected to QGroundControl.

More about the NSH shell: https://dev.px4.io/v1.9.0/en/debug/system_console.html

Help

? or help Shows a list of commands and Build-in Apps:



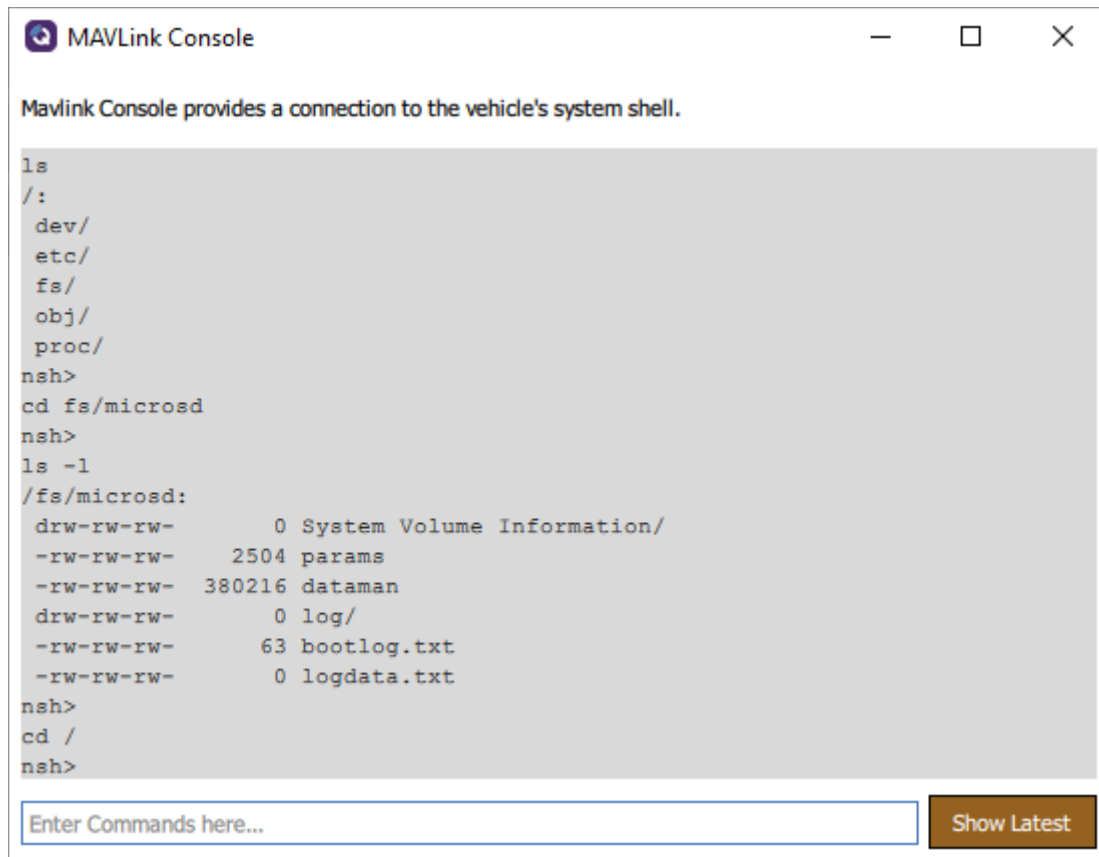
Those shell commands maybe helpful just for information or for troubleshooting. Please execute commands only if you know what you do. Refer to the PX4 Autopilot documentation and QGroundControl manual.

Following commands can be executed without problems, do not change anything and are only used to show some information.

How to use custom command **typhoon_bind** please refer to the Thunderbird user manual.

File system

ls	List files or directories
cd fs/microsd	Change directory to list the files on micro SD card
ls -l	List the files or directories on micro SD card
cd /	Change directory to root directory



```
MAVLink Console
Mavlink Console provides a connection to the vehicle's system shell.

ls
/:
dev/
etc/
fs/
obj/
proc/
nsh>
cd fs/microsd
nsh>
ls -l
/fs/microsd:
drw-rw-rw- 0 System Volume Information/
-rw-rw-rw- 2504 params
-rw-rw-rw- 380216 dataman
drw-rw-rw- 0 log/
-rw-rw-rw- 63 bootlog.txt
-rw-rw-rw- 0 logdata.txt
nsh>
cd /
nsh>
```

With the commands above you can check if you have an SD card inserted and if it is readable.

To download flight logs (ULOG files: *.ulg) use QGroundControl **Analyze > Log Download**. Click on **Refresh** to see updated list of ULOG files.

Devices

ls List directories in root directory
cd dev Change directory to 'dev'
ls List devices of the drone



The screenshot shows a window titled "MAVLink Console" with standard window controls (minimize, maximize, close). Below the title bar, a message states: "Mavlink Console provides a connection to the vehicle's system shell." The main area is a terminal window with a light gray background. It shows the following commands and output:

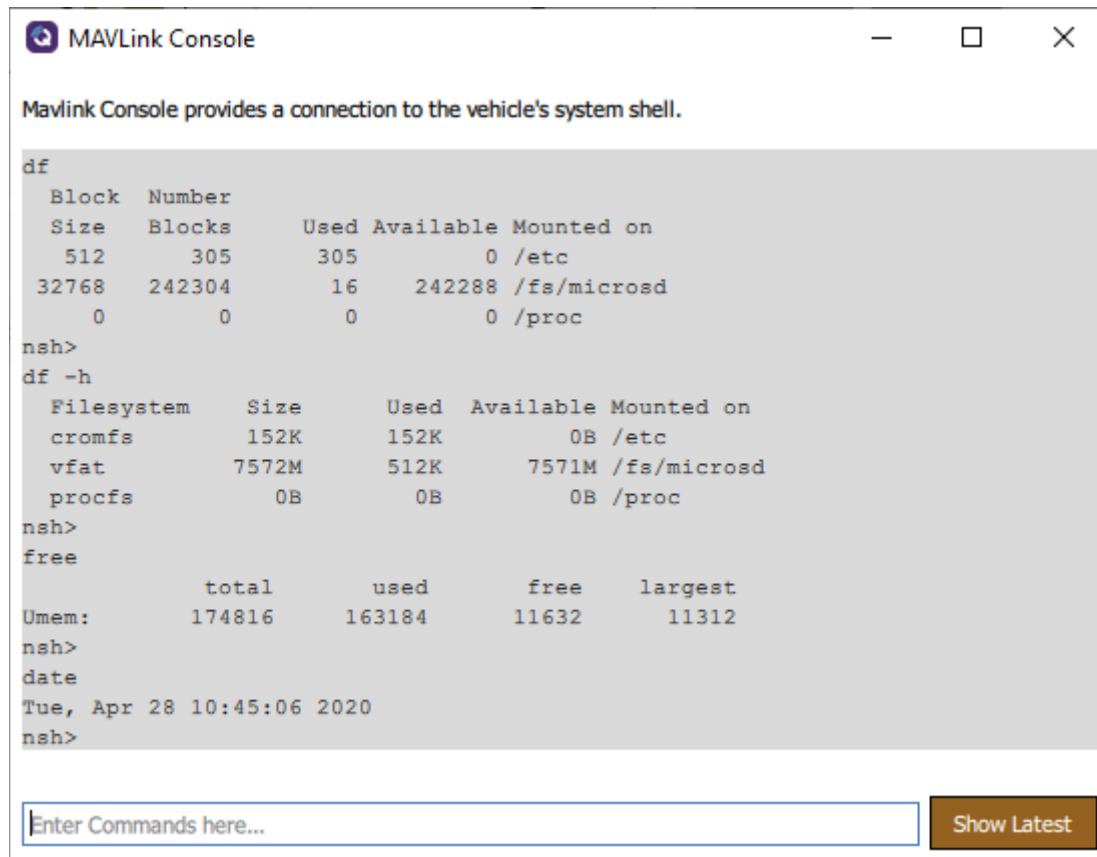
```
nsh>
ls
/:
 dev/
 etc/
 fs/
 obj/
 proc/
nsh>
cd dev
nsh>
ls
/dev:
 accel0
 adc0
 baro0
 console_buf
 gyro0
 hmc5883_ext
 mag0
 mmcsd0
 mpu6000
 ms5611_int
 null
 pipe0
 pipe1
 pwm_output0
 px4fmu
 tap_esc
 ttyACM0
 ttyS0
 ttyS1
 ttyS2
 ttyS3
nsh>
```

At the bottom of the terminal area, there is a text input field labeled "Enter Commands here..." and a button labeled "Show Latest".

mpu6000 info Learning more about IMU

System

help	List commands and apps
df	Show memory sizes
df -h	Show memory sizes in 'human readable' format
free	Show free/used memory
date	Show system time



MAVLink Console

Mavlink Console provides a connection to the vehicle's system shell.

```
df
  Block  Number
  Size   Blocks    Used Available Mounted on
  512      305     305         0 /etc
 32768  242304      16    242288 /fs/microsd
    0         0         0         0 /proc

nsh>
df -h
  Filesystem    Size      Used Available Mounted on
  cromfs        152K     152K         0B /etc
  vfat          7572M     512K     7571M /fs/microsd
  procfs         0B         0B         0B /proc

nsh>
free
             total        used        free      largest
Umem:       174816     163184     11632       11312

nsh>
date
Tue, Apr 28 10:45:06 2020

nsh>
```

Enter Commands here... Show Latest

sensors status	Information about sensors
gps status	Information about GPS module (dev/ttyS0, 115200 baud)
rc_input status	Information about GPS module (dev/ttyS3)

and many more...

Note: Channel settings on ST16 are stored to the related models. If you want to reset Channel settings, create a new model from scratch.

[illegible]

Cheat sheet



Proposal for brief description to print it.

Flight mode switch setting (fill table with your own settings)

FlightMode	Switch	Position
Position (Angle mode)	Flight mode switch S4	middle
Altitude (GNSS support off)	Flight mode switch S4	upper
Stabilized (Manual)	Aux switch B2	upper
Rattitude	Aux switch B2	lower
Acro		
Mission	Flight mode switch S4	lower
RTH	Pan mode switch S2	lower

Power on/off

Power on: Hold power button on drone for 8sec until LEDs are on.
The Speed Slider must be set to **"Rabbit"**!

Power off: Remove battery

Motor Start / Stop

Hold throttle stick in the lower right corner

Flying

- ✓ Check location (rules and regulations)
- ✓ Free space for GPS signals, no potential reflectors to avoid multi-path reception
- ✓ Drone battery full, ST16 battery full
- ✓ **Check battery voltage during flight**

Bind mode: nsh shell command "typhoon_bind start"